

Literature status note for arXiv:1209.0055

Two-dimensional Mazur–Ulam-property subcase

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Status

`literature_implied_answer` (partial subcase).

Source problem

The source paper is Dongni Tan and Rui Liu, *A Note on The Mazur-Ulam Property of Almost-CL-spaces*, arXiv:1209.0055. In the introduction they recall Tingley’s isometric extension problem as Problem 1:

Let X and Y be normed spaces with unit spheres S_X and S_Y . Assume that $T : S_X \rightarrow S_Y$ is a surjective isometry. Does there exist a linear isometry $\tilde{T} : X \rightarrow Y$ such that $\tilde{T}|_{S_X} = T$?

They then formulate the Mazur–Ulam property and state that the problem is still open in general, even in two dimensions. The same paper proves positive results for spaces with the (T) -property, almost-CL spaces admitting smooth points, one hexagonal two-dimensional norm, and some stability constructions; those same-paper results are not the basis of this status note.

Supporting theorem

The supporting paper is Taras Banach, *Every 2-dimensional Banach space has the Mazur–Ulam property*, arXiv:2103.09268; Linear Algebra Appl. 632 (2022), 268–280.

Its abstract states that every isometry between the unit spheres of 2-dimensional Banach spaces extends to a linear isometry and that this resolves Tingley’s problem in the class of 2-dimensional Banach spaces. The main theorem, Theorem 1, states:

Every 2-dimensional Banach space has the Mazur–Ulam property.

This theorem answers the two-dimensional subcase that Tan–Liu described as still open in 2012.

Scope

This packet does not claim a full solution of Tingley’s problem or of the general Cheng–Dong Mazur–Ulam question asking whether every Banach space has the Mazur–Ulam property. The supporting

theorem settles only the 2-dimensional case. Later literature still describes the problem as open in finite dimensions at least three, while citing Banach for the 2-dimensional positive answer.

The relation is an agent-identified implication. Banach's paper cites Cheng–Dong for the Mazur–Ulam-property formulation and explicitly solves the named Tingley problem in dimension two, but it does not appear to cite Tan–Liu's almost-CL paper directly.

Search evidence

The run indexes were searched for arXiv:1209.0055 and the keywords “Mazur-Ulam property”, “Tingley”, “almost-CL”, “MUP”, and “sphere isometry”. No exact duplicate packet for arXiv:1209.0055 was found, although nearby Tingley/MUP packets already exist for other subclasses. External web search on 2026-06-29 and local parsed-source inspection identified arXiv:2103.09268 as the decisive answer for the two-dimensional subcase.